Challenge 1: Clinical Trial Radar (Trial Landscape & Feasibility Explorer)

Area: Pharma
Potential customer: Clinical Ops, Feasibility/Study Start-Up, Country & Site Strategy, CROs, Medical Affairs

Problem statement:
Clinical trials fail or delay most often because of recruitment and feasibility assumptions made too early. Teams need to decide countries/sites and timelines quickly, but the information is scattered and hard to compare. People still build feasibility in spreadsheets, copying trial details manually from multiple sources. That means decisions can be inconsistent across teams, and reality hits late: regions are saturated, competing trials are recruiting the same patients, or expected timelines were unrealistic. This creates high cost, slow execution, and rework (protocol changes, country swaps, delayed activation).

Vision statement:
Provide a simple "radar" that shows what's happening in the clinical trial world for a disease and phase: where trials are happening, how crowded recruitment is, what the trend looks like over time, and what typical cycle times are. The goal is not to replace clinical judgment, but to give teams a fast, factual baseline so feasibility and planning decisions become faster, more consistent, and less risky.

Product Overview:

Business overview: Clinical Trial Radar reduces feasibility cycle time and helps teams avoid late-stage surprises by turning public registry data into decision-ready insights. It helps stakeholders align earlier with a shared "single view" of the landscape and recruitment competition.
Product overview: The tool ingests trial registry data, normalizes disease terms and geographies, and produces trends, maps, and "crowdedness" signals (active recruiting competition). Optional NLP surfaces common eligibility and endpoints. A one-click "Feasibility Snapshot" export makes it usable for governance and approvals.

MVP scope (example)

- Search by disease + phase
- Trend chart: recruiting vs completed over time
- Map: trials by country (optionally by city)
- Crowdedness metric: active recruiting trials in region
- Table drill-down to trial records
- Export snapshot PDF/slide

KPIs

- Time saved for feasibility research
- Coverage: % of relevant trials captured for query

- Consistency: standardized feasibility output
- Proxy: fewer feasibility-driven changes after kickoff

Useful links (open data + docs)

Primary trial data

- ClinicalTrials.gov API v2: https://clinicaltrials.gov/data-api/about-api
- ClinicalTrials.gov bulk downloads: https://clinicaltrials.gov/data-download
- AACT (SQL-ready ClinicalTrials dataset): https://aact.ctti-clinicaltrials.org/
- AACT docs: https://aact.ctti-clinicaltrials.org/documentation

Additional trial registries (open portals / datasets)

- EU Clinical Trials Register (public search): https://www.clinicaltrialsregister.eu/
- ISRCTN registry (public search): https://www.isrctn.com/

Terminology + enrichment

- MeSH: https://www.nlm.nih.gov/mesh/meshhome.html
- MeSH downloads: https://www.nlm.nih.gov/databases/download/mesh.html
- PubMed E-utilities: https://www.ncbi.nlm.nih.gov/books/NBK25501/
- OpenAlex: https://openalex.org/
- NIH RePORTER: https://api.reporter.nih.gov/

Optional feasibility proxies

- WHO GHO: https://www.who.int/data/gho
- World Bank: https://data.worldbank.org/

Challenge 2: Pharmacovigilance Signal Triage Copilot (Early Safety Signals + Prioritization)

Area: Pharma
Potential customer: PV/Patient Safety, Medical Affairs, Regulators, PV vendors

Problem statement:
Safety teams receive thousands to millions of side-effect reports. These reports are messy: drug names vary (brand/generic/spelling), reactions are described differently, and cases can be duplicated or incomplete. Humans must still triage, normalize, and compile evidence before experts can assess signals. When triage is slow, dangerous patterns can be missed or seen late; when triage is overwhelmed, teams spend time on noise rather than real risk. The core need is faster, explainable prioritization, not automated causality.

Vision statement:

Create a tool that works like a “smoke detector” for safety: it flags unusual drug–event patterns early, explains why they were flagged, and prepares a review-ready evidence pack. Experts stay in control, but they start from a ranked list with context instead of a raw pile of reports.

Product Overview:

Business overview: The copilot reduces manual triage burden and improves consistency, helping PV teams focus on the highest-risk signals sooner. It supports audit-friendly workflows by producing standardized evidence summaries.

Product overview: The system ingests open adverse event data, normalizes drug names/reaction terms, computes disproportionality metrics (PRR/ROR) and trends, and identifies rising signals. It outputs explainable “signal packets” with counts, baseline comparisons, example cases, and charts.

MVP scope (example)

- Select drug + time window
- Top drug–event signals with PRR/ROR
- Trend vs baseline
- Drill-down into example reports
- Export signal packet
- Simple ranking (seriousness + growth)

KPIs

- Time-to-triage reduced
- Signal packets produced per analyst per day
- Earlier detection proxy (time from emergence to review)
- Standardization score (same evidence format)

Useful links (open data + docs)

Primary PV data

- openFDA FAERS API: https://open.fda.gov/apis/drug/event/
- FAERS background: https://www.fda.gov/drugs/questions-and-answers-fdas-adverse-event-reporting-system-faers/fda-adverse-event-reporting-system-faers-public-dashboard

Drug normalization + open drug data

- RxNorm: https://lhncbc.nlm.nih.gov/RxNorm/
- DrugCentral (open drug database): https://drugcentral.org/
- ChEMBL (bioactivity + compounds): https://www.ebi.ac.uk/chembl/
- Open Targets (target-disease associations): https://platform.opentargets.org/

Optional safety-related sources

- FDA Recalls (open): https://open.fda.gov/apis/food/enforcement/ *(useful pattern for "alerts/recalls" style)*
- PubMed E-utilities (literature): https://www.ncbi.nlm.nih.gov/books/NBK25501/

Challenge 3: Omnichannel Engagement Optimizer (HCP Journey + Next Best Action, Compliant)

Area: Pharma
Potential customer: Commercial Excellence, Brand, Omnichannel/Digital, Field leadership, CRM, Medical/MLR

Problem statement:
Pharma engages HCPs across many channels, but data and planning are siloed: email teams optimize email, reps optimize visits, events teams run webinars, and nobody has a unified view of the journey. Without orchestration, HCPs get duplicate touchpoints or fatigue, and teams cannot clearly answer what drives meaningful engagement. On top of that, compliance constraints (consent, approved content, frequency caps) limit experimentation. The result is wasted spend, inconsistent experiences, and weak ROI proof.

Vision statement:
Create a compliant orchestration cockpit that shows HCP journeys end-to-end, recommends next best actions under governance constraints, and tracks measurable outcomes. The goal is "right message, right channel, right timing" with transparency, frequency control, and audit-friendly logic.

Product Overview:

Business overview: The tool improves engagement efficiency, reduces duplication, and strengthens ROI measurement by aligning field + digital around one journey view. It also reduces compliance risk by enforcing governance rules in the recommendation flow.

Product overview: Using synthetic or anonymized data, the system segments HCPs, visualizes multi-channel activity, and recommends actions using rules or lightweight propensity scoring. A compliance layer enforces approved content, opt-in/consent simulation, and frequency caps. Outputs include weekly action plans and KPI dashboards.

MVP scope (example)

- Synthetic HCP profiles + interactions
- Journey segmentation (aware → engaged → active)
- Channel dashboards
- Rule-based NBA with constraints

- Compliance library + frequency caps
- Weekly plan export

KPIs

- Engagement lift proxy (per segment)
- Touches per meaningful action (efficiency proxy)
- Fatigue indicators (contact distribution / opt-out proxy)
- Compliance coverage (% compliant recommendations)
- Cost per engagement milestone (demo ROI)

Useful links (open data + docs)

Synthetic + modeling

- Synthea: https://synthea.mitre.org/
- OMOP CDM: https://www.ohdsi.org/data-standardization/the-common-data-model/
- Kaggle marketing datasets: https://www.kaggle.com/datasets?search=marketing+campaign
- Lightweight MMM: https://github.com/google/lightweight_mmm

Public healthcare professional datasets (use carefully)

- US NPI Registry (HCP identifiers): https://npiregistry.cms.hhs.gov/
- CMS Open Payments (financial relationships): https://openpaymentsdata.cms.gov/

Compliance context

- FDA promotion basics: https://www.fda.gov/drugs/prescription-drug-advertising

Challenge 4: Intelligent Context Router (Prevent "Context Window Overflow" in Patient Summaries)

Area: Hospital
Potential customer: Hospitals (informatics, EHR teams), digital health vendors

Problem statement:
Hospitals want LLM summaries, but long patient histories produce huge FHIR bundles that exceed token limits. Naive chunking/truncation can remove critical context and create unsafe summaries. The missing capability is an intelligent layer that selects, compresses, and prioritizes clinically relevant data for a specific question while preserving traceability to original records.

Vision statement:
Build a safe routing layer that generates a compact, structured "context pack" for the LLM, preserving critical facts and preventing information loss for long histories.

Product Overview:

Business overview: The router reduces clinician time spent navigating history while maintaining safety and traceability. It enables scalable summarization without sending entire records to the LLM.

Product overview: The tool ingests large FHIR bundles, ranks resources by relevance using SQL/embeddings/rules, compresses into structured sections, and sends only the prioritized context to the summarization model. Outputs include the summary plus a "source map" to the underlying FHIR IDs.

MVP scope

- Load synthetic FHIR bundle
- 3 clinical question types
- Resource ranking + selection
- Structured context pack + summary
- Traceability to FHIR IDs
- Token reduction report

KPIs

- Token reduction %
- Recall of critical facts
- Summary completeness rubric
- Latency per query

Useful links (open data + docs)

- HL7 FHIR: https://www.hl7.org/fhir/
- SMART on FHIR: https://smarthealthit.org/
- Synthea: https://synthea.mitre.org/
- OMOP CDM: https://www.ohdsi.org/data-standardization/the-common-data-model/
- MIMIC-IV (open, requires credentialing): https://physionet.org/content/mimiciv/

Challenge 5: Temporal Drift Analyzer (Detecting Drift in Synthetic EHR Data)

Area: Hospital
Potential customer: Hospital AI teams, academia, synthetic data vendors

Problem statement:
Healthcare changes over time, but synthetic datasets often don't. When models trained on static

synthetic data meet real-world shifts (new codes, new diseases, new protocols), performance degrades. Teams lack an easy way to measure if "future-like" data differs in meaningful ways and where those changes concentrate.

Vision statement:
Create a drift analyzer that detects and explains temporal change across domains (diagnosis, meds, LOS, utilization), enabling better model monitoring and better synthetic data generation targets.

Product Overview:

Business overview: The analyzer reduces deployment risk by making drift visible and measurable before it causes clinical harm or operational failure. It supports governance and monitoring plans.

Product overview: It compares baseline vs future datasets, calculates drift metrics across features, ranks top drift drivers, and generates an interpretable drift report with charts and "what changed" explanations.

MVP scope

- Two datasets (baseline + future)
- Drift metrics dashboard
- Domain-level drift views
- Top drift drivers list
- Export report
- Validate with injected shifts

KPIs

- Drift detection precision/recall
- % feature coverage
- Time to identify drift drivers
- Interpretability (human review)

Useful links (open data + docs)

- Synthea: https://synthea.mitre.org/
- MIMIC-IV (open, credentialed): https://physionet.org/content/mimiciv/
- eICU (open, credentialed): https://physionet.org/content/eicu-crd/
- TensorFlow Data Validation (drift/data checks): https://www.tensorflow.org/tfx/guide/tfdv
- Concept drift background: https://en.wikipedia.org/wiki/Concept_drift

Challenge 6: Semantic Alarm Deduplication Engine (Reducing ICU Alarm Fatigue)

Area: Hospital
Potential customer: Hospitals (ICU, clinical engineering, IT), patient safety groups

Problem statement:
Clinicians get flooded with alarms, many of which are duplicates from different systems describing the same event. Simple rules miss semantic duplicates, so noise stays high and fatigue increases. The need is a real-time dedup layer that understands meaning, not just identical text, and consolidates redundant alarms safely.

Vision statement:
Create a semantic deduplication engine that groups redundant alarms into a single actionable notification, reducing noise while preserving safety and traceability.

Product Overview:

Business overview: Reducing redundant alarms improves response quality and reduces fatigue risk while keeping clinicians informed. It helps hospitals improve patient safety and operational efficiency.

Product overview: The system consumes alert streams, embeds messages, clusters semantically similar alerts within a short time window, and outputs a unified alert with source traceability and priority logic.

MVP scope

- Simulated multi-source alert stream
- Embedding similarity + clustering
- Unified alert output
- Threshold configuration
- Evaluation dashboard
- False merge checks

KPIs

- Dedup rate %
- False merge rate
- Time to consolidated alert
- Alerts/hour reduction proxy

Useful links (open data + docs)

- Alarm fatigue primer (AHRQ): https://psnet.ahrq.gov/primer/alarm-fatigue
- Sentence-BERT: https://www.sbert.net/

- spaCy: https://spacy.io/
- PhysioNet datasets (credentialed, for realistic signals): https://physionet.org/

Challenge 7: AI Platform for Medical Quality Assurance & Automated Regulatory Review

Area: MedTech
Potential customer: Public sector, healthcare authorities, medical associations, large providers

Problem statement:
Regulatory and QA teams must review large volumes of documentation and evidence, but manual review is slow, inconsistent, and costly. Important checks like completeness, requirements matching, and cross-document consistency often consume most time. The consequence is long cycle times, variable decisions, and low transparency. There's a need for a platform that automates routine checks and supports reviewers with traceable summaries.

Vision statement:
Build a centralized platform that speeds up review while improving consistency and transparency, using explainable outputs and governance suitable for high-risk workflows.

Product Overview:

Business overview: The platform reduces review cycle time and improves consistency by automating repetitive validation and structuring the decision-support process. Reviewers spend time on judgment instead of paperwork.

Product overview: The system ingests submissions, classifies them, checks completeness and requirements alignment, and uses RAG over guidance documents to support reviewer questions. It outputs an audit-friendly review pack with traceability.

MVP scope

- Upload docs + metadata
- Submission classification
- Completeness checklist
- RAG search over guidance corpus
- Structured review summary generation
- Export decision pack

KPIs

- Review time reduction
- Completeness detection accuracy
- Consistency proxy (agreement rate)
- Audit pack completeness %

Useful links (open data + docs)

Regulatory + safety open datasets

- FDA 510(k) database: https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm
- FDA recalls (device): https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfres/res.cfm
- FDA MAUDE adverse event reports (device): https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfmaude/search.cfm
- WHO AI governance: https://www.who.int/publications/i/item/9789240029200
- RAG overview: https://learn.microsoft.com/en-us/azure/ai-services/openai/concepts/retrieval-augmented-generation

Challenge 8: Medical Device Cybersecurity-by-Design Platform (SBOM, Patching, Threat Monitoring)

Area: MedTech
Potential customer: Device manufacturers, hospitals (biomed + IT), public procurement

Problem statement:
Connected medical devices introduce cyber risk and compliance pressure. Many organizations don't have complete visibility into device inventory, software components, known vulnerabilities, and patch status. Response is often reactive and documentation-heavy. The consequence is increased exposure, slower remediation, and difficult audits/procurement.

Vision statement:
Make device cybersecurity continuous and lifecycle-based: always know what you have, what it contains, what's vulnerable, and what needs patching—plus generate compliance-ready evidence automatically.

Product Overview:

Business overview: The platform reduces risk and operational disruption by prioritizing vulnerabilities, accelerating remediation, and standardizing evidence packs for audits and procurement.

Product overview: It manages SBOMs, matches components to CVEs, scores risk using public exploit intelligence, tracks patch status, and presents hospital/manufacturer dashboards. It exports compliance packs and mitigation guidance.

MVP scope

- Inventory + SBOM upload
- CVE matching
- Risk scoring + prioritization

- Patch status tracking
- Dashboard (top risks + actions)
- Export evidence pack

KPIs

- MTTD exposure
- MTTR remediation
- Patch compliance %
- SBOM coverage %

Useful links (open data + docs)

- FDA cybersecurity guidance: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/cybersecurity-medical-devices-quality-management-system-considerations-and-content-premarket
- NVD (CVE database): https://nvd.nist.gov/
- CISA KEV (known exploited vulnerabilities): https://www.cisa.gov/known-exploited-vulnerabilities-catalog
- NIST CSF: https://www.nist.gov/cyberframework
- NTIA SBOM resources: https://www.ntia.gov/sbom
- SPDX (SBOM standard): https://spdx.dev/
- CycloneDX (SBOM standard): https://cyclonedx.org/
- Device cybersecurity review: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10222766/